

Heavy Looks, Slower Moves: Effects of Physical and Visual Object Weight on Pointing in Virtual Reality

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Abstract

In Virtual Reality (VR), users frequently interact through pointing movements. The accuracy and speed of such goal-directed movements depend on their estimated energy and time costs. In VR, vision is stimulated separately from tactile and kinesthetic sensory modalities. For instance, a virtual pointing device can be visually presented as heavy while being physically lightweight. Yet, it is unknown how visual and physical weight cues contribute to our strategy to optimize movements in terms of accuracy and speed. In a study with 32 participants, we found physical weight (additional weight of the controller) increases precision and movement time when pointing in VR. Interestingly, we found that visually conveyed weight (additional volume at the controller's 3d model) also increases movement time, suggesting that adjusting the appearance of virtually held objects can affect the performance of pointing movements. We interpret our findings in light of motor-cognitive models and discuss implications for VR designers.

CCS Concepts

• **Human-centered computing** → **Haptic devices; Virtual reality; Pointing.**

Keywords

Virtual reality, Pointing, Speed, Accuracy, Ray cast, Goal-directed movement, Weight illusion, Weight simulation

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1 Introduction

In Virtual Reality (VR), object manipulation, teleportation, and menu item selection are essential elements for user engagement. One of the most common approaches for facilitating these interactions is ray casting, in which users perform pointing movements to aim at objects, menu items, or teleportation areas [26, 33]. In real-world settings, goal-directed movements, such as pointing, are planned and executed to optimize speed, accuracy, and energy expenditure [10]. Factors, such as the physical weight of a handheld object, can affect the expected energy costs and thereby influence the movement strategy [6]. In VR, sensory cues from different modalities that affect the perceived weight — such as vision, haptics, and tactile feedback — can be modified. For example, while a physical controller may be lightweight, it can be visually presented to the user as a heavier object, by upscaling its dimension [14], depicting a denser material [36], or visually replacing it with a heavy object of familiar weight [41]. The separation between visual and physical cues in VR raises questions about how conflicting information from different modalities impact movement performance. Specifically, it is unknown if an object's visually implied weight can influence how users approach and execute goal-directed tasks within VR, even if the object's physical weight remains unchanged. Additionally, weight interfaces are capable of adjusting the physical weight of the handheld controller (e.g. [7, 17, 20–22, 30, 40]), to enhance realism by conveying the weight of virtual objects. However, it is unknown, whether such adjustments of the physical weight

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could be used to manipulate the movement performance of VR users when precision and efficiency are crucial.

To understand how the physical and the visual weight of hand-held objects affect users' pointing movements in VR, we conducted a study with 32 participants, in which we manipulated both, the physical weight of the VR controller and the visually conveyed weight of its representation in VR. We found that participants moved more accurately, but slower when using a physically heavier pointing device. Critically, regardless of the physical weight of the pointing device, participants performed their movements faster when moving the lighter-looking virtual representation of the pointing device. Our results indicate that users adapt their strategy to perform pointing movements based on the virtual weight implied by the appearance of the virtual object. Furthermore, they show that physically increasing the weight of the input device could amplify the precision of the users' movements.

2 Related Work

2.1 Pointing in Virtual Reality

Pointing is among the most widely used techniques for target selection in VR. Typically, a ray is cast from the VR controller, imitating a laser pointer to allow users to select objects, menu items, or teleportation areas [26, 33]. Although this technique is efficient, challenges arise when users point at distant objects or in cluttered environments. Physiological tremors can disturb the spatially tracked input device [32, 39]. Similarly, discrete user inputs, such as button presses, can cause the selection point to shift unexpectedly—a phenomenon known as the Heisenberg Effect of Spatial Interaction [42]. The theoretical underpinnings of pointing as a goal-directed movement are well-explained by motor-cognitive models like the "Optimized Initial Impulse Model", which was proposed by Meyer et al. [29] to explain the logarithmic relationship between movement time and spatial accuracy (Fitts' law). In line with pioneering work on limb control [43], the model indicates that aimed movements consist of a primary ballistic submovement that brings the limb close to the target and a secondary corrective submovement. It was extended by Elliott et al. [9], who assume that humans undershoot the target in the primary submovement to avoid the time and energy costs associated with changing the direction after an overshoot. The authors derived a model according to which humans balance speed, accuracy, and energy use when performing goal-directed movements [8]. In support of this model, Burkitt et al. [6] found the weight of a stylus to impact the performance of aimed tapping movements on a flat surface. Participants undershoot to a greater extent when using a 243 g stylus as compared to a 36 g stylus. The authors hypothesize that this arose from additional weight being associated with higher overshoot costs. Their findings suggest that the perceived weight of a handheld object is taken into account when balancing speed accuracy and energy consumption to plan an aimed movement. However, it is unclear how these findings can be applied to pointing movements in VR, where the senses through which we perceive weight are stimulated separately.

2.2 Weight perception and simulation in VR

We perceive the weight of objects through kinesthetic, haptic, and visual signals. Kinesthetic perception stems from the force we apply

to hold or stabilize an object against gravity, sensed through our muscles, skin, and joints [38]. This includes resisting an object's rotational dynamics. Consequently, an object's perceived heaviness also depends on its internal weight distribution [2] and the applied grip [27]. Tactile perception, in contrast, is driven by the sensation of forces acting on the skin, such as the skin being pulled down due to an object's weight [28]. Visual perception is also critical, introducing biases based on prior experiences and expectations. For instance, phenomena like the size-weight illusion demonstrate how larger objects are perceived as lighter than equally weighted smaller ones due to a learned relationship between weight and size [5]. Similarly, object identification influences perception, as seen in the "golf ball illusion," where familiarity with an object can establish preconceived notions about its weight [11].

VR allows for the manipulation of these sensory weight cues. In standard setups, these are at odds with each other, as, even when the sense of vision perceives a heavy object to be held, the kinesthetic system perceives the light weight of the VR controller. Altering the visual appearance of the handheld virtual object creates different anticipations of weight [18]. This can affect a virtual object's perceived heaviness, as demonstrated by Maehigashi et al. [25] and Shimamura et al. [36], who consistently found that in VR, objects with heavier-looking materials were also perceived as heavier.

To perceive kinesthetic and tactile forces, haptic devices that adjust their own mass [19, 22, 40] or generate kinesthetic forces through motors [1, 15, 16] can be employed. When moving virtual objects, kinesthetic weight properties can also be altered by manipulating the control/display ratio [34, 35, 44]. Tactile cues can be adjusted through interfaces that apply localized pressure to the skin, mimicking the sensation of weight pulling downward [12, 23, 37].

In sum, VR systems decouple the sensory pathways through which we perceive weight. It is, however, unknown how the distinction of physical and visual weight cues affect our movements and whether it can be exploited to manipulate the performance in pointing tasks.

3 Method

3.1 Study Design

We conducted a study to investigate how the physical and visually conveyed weight of handheld objects affect users' pointing movements in VR. In the study, participants used two different weighted pointing devices to point at targets, while in the virtual world either the weighted or unweighted device was displayed, matching or not matching its real-world counterpart. We used a within-subjects design, with the two independent variables PHYSICAL WEIGHT and VISUAL WEIGHT, both with the two levels 400 g and 800 g. These weight levels were chosen based on the weight of the devices' components and the feasibility of performing all trials comfortably with them. PHYSICAL WEIGHT refers to the physical weight of the pointing device used. VISUAL WEIGHT refers to the 3D replications of the two pointing devices representing the pointing device in the virtual world. To prevent order effects, we counterbalanced the order of the resulting four conditions using a 4 × 4 Latin square.

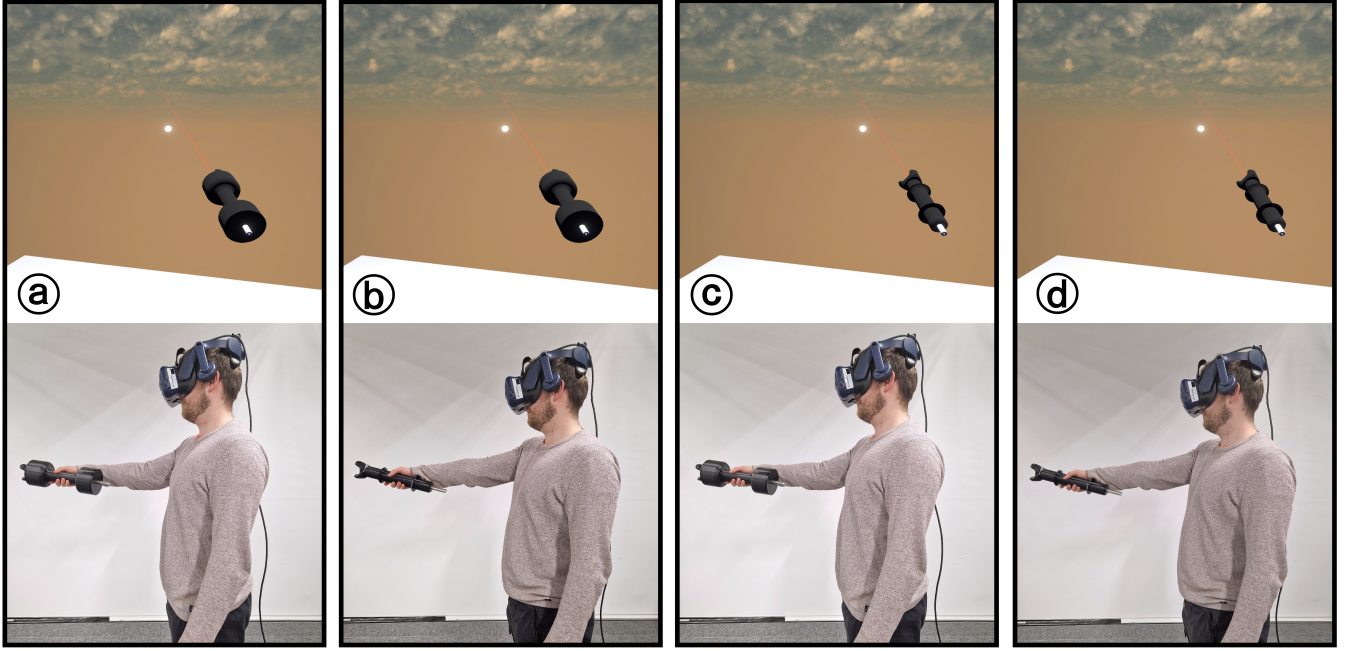


Figure 1: Participant with their respective handheld device (bottom) and their VR view (top) in the four experimental conditions (a: 800 g physical with 800 g visual weight, b: 400 g physical with 800 g visual weight, c: 800 g physical with 400 g visual weight, d: 400 g physical with 400 g visual weight).

3.2 Task

3.2.1 Priming Task. An object’s expected weight depends on individual experiences [11]. Hence, to prevent inconsistent expectations of the visual weight among participants, each main task was preceded by a priming task outside of VR, in which the participants used the pointing device they would see as a visual representation in the following VR task, to point at targets in the physical space. In the priming task, 38 targets (white circles) were subsequently projected onto the wall in our laboratory. These projections were initiated by a slide show controlled by the experimenter, who advanced to the next slide when the target was hit by the point of light emitting from the pointing device’s integrated laser pointer. The priming task did not involve any collection of data.

3.2.2 Main Task. In the main task, participants were instructed to aim at 80 targets in VR using the current condition’s pointing device. They were asked to perform the movements with a slightly outstretched arm to maintain consistency between participants while ensuring the virtual pointing device was within the field of view. Following standard selection techniques in VR [33], a visible red ray of light emits from the pointing device, mimicking a laser pointer. The targets, presented as white glowing spheres, consecutively spawn on a hemispherical sky dome. Only one target was presented at a time. Upon continuously pointing at the target for 200 ms, it disappeared and the next target spawned on the sky dome. The spawn position of the target was randomly generated, with the constraint that it was positioned two meters away from the previous target and five meters away from the VR headset. After

the 80th target, the task was complete. Of the 80 trials, the first 10 were practice trials, which were not part of the analysis.

3.3 Measures

3.3.1 Task Completion Time. We assessed the task completion time as the time taken by participants to successfully aim at the new target after its appearance, i.e., the time in milliseconds between the appearance of a target and its destruction, including the dwell time of 200 ms.

3.3.2 Precision. We used the deviation of the light ray’s covered path from the optimal path to quantify the precision of the participant’s movements. We assessed the length of the path traveled by the light ray on the surface of the sky dome to reach each target, where a shorter path length indicates less deviation from the optimal path and therefore greater precision. The optimal path between two targets that were positioned at a Euclidean distance of two meters on the sky dome is approximately $2r \arcsin\left(\frac{d}{2r}\right) \approx 2.013$, where $r = 5$ m is the radius of the sky dome and d is the Euclidean distance of the two targets, which is always 2 m. The displacement of the light ray was calculated using the Haversine formula. During the task, the light ray’s current and previous position on the sky dome was determined continuously (at 30 Hz), and the distance between these positions was calculated. By summing the displacements obtained in this way, we retrieved the complete trajectory drawn by the participant in each trial.

3.3.3 Questionnaires. We assessed realism after each condition using the Reality Judgment and Presence Questionnaire [3], which



Figure 2: The 400 g (left) and 800 g (right) pointing devices and their respective technical drawings.

contains 18 items, scored on a Likert scale ranging from 0 ("not at all") to 10 ("absolutely"). Furthermore, we obtained the participants' perceived physical exertion through the psychophysical Borg Rating of Perceived Exertion scale [4], which ranges from 6 (perceiving "no exertion at all") to 20 (perceiving "maximal exertion").

3.4 Stimuli

The physical pointing devices and their proportions are shown in Figure 2. Both devices consist of an elongated cylindrical housing featuring a handle in its center. The 800 g device has supplementary mass and volume on each end, resembling a weightlifting dumbbell to evoke the notion of added weight and to create the expectation to be heavier than the 400 g device. Apart from the additional weight and volume on the sides, it is identical to the 400 g device. The devices' housings were 3D-printed. Both devices have a laser pointer attached to one end and an HTC Vive tracker (3.0) attached to the other. To reach the 400 g or 800 g weight, and to balance the weight distribution, sandbags were placed in empty space at the controller's end. To create the visual stimuli, we used the 3D models of the devices we created for 3D printing, assigned them a black material, and combined them with models of the Vive tracker and the laser pointer, yielding virtual replicas of the physical pointing devices.

3.5 Apparatus

The VR environment, consisting of a white platform in a desert valley and a skybox, was created using the game engine Unity (2021.3.15f1). The application ran on a desktop PC equipped with an Intel i7-8750H Processor, 16GB RAM, and an NVIDIA GeForce GTX 1060. Participants experienced the virtual environment through an HTC Vive Pro. To track the movements of the participants, we used SteamVR base stations 2.0.

3.6 Participants

We recruited 33 participants through institutional mailing lists. One participant with color vision deficiency was excluded, resulting in a sample size of 32 (23 male, 9 female). The participants' ages ranged from 19 to 34 ($M = 25.16$, $SD = 3.35$). Participants were compensated with credit points for their study program. Four participants reported using VR regularly, sixteen had used it a few times, and the remaining twelve had never used VR.

3.7 Procedure

After signing an informed consent form, the participants were provided with an explanation of the task and the functionality of the equipment. They were instructed to stand on a stationary point marked on the floor during the task. To encourage temporal efficiency, participants were informed that the faster they completed the task, the higher their overall score would be. In each of the four conditions, participants conducted the priming and the main task. After each condition, the participants completed the Reality Judgment and Presence Questionnaire and rated their perceived exertion using the BORG scale. After finishing the four conditions, participants filled out a demographic questionnaire.

4 Results

We excluded the first 10 trials of each round, as these were practice trials. We further removed trials with an extreme task completion time of over 3000 ms, to exclude data points in which the participant was distracted. Ultimately, 97.73% of all experimental trials were included in our analysis. We employed a 2 (PHYSICAL WEIGHT: 400 g vs. 800 g) $\times$ 2 (VISUAL WEIGHT: 400 g vs. 800 g) repeated measures analysis of variance (ANOVA) for all of our data.

4.1 Task Completion Time

A 2×2 ANOVA of the task completion time revealed a main effect of PHYSICAL WEIGHT, $F(1, 31) = 16.314$, $p < .001$, $\eta_p^2 = .345$, indicating that participants took longer to aim when using the 800 g device compared to the 400 g device. There was also a main effect of VISUAL WEIGHT, $F(1, 31) = 7.945$, $p = .008$, $\eta_p^2 = .204$, showing that participants took longer to aim when the 800 g model was displayed in VR compared to the virtual 400 g model. However, there was no interaction effect of PHYSICAL WEIGHT $\times$ VISUAL WEIGHT, $F(1, 31) = .353$, $p = .531$, $\eta_p^2 = .013$. Figure 3 depicts the mean values of the task completion time.

To test for any fatigue effect over the trials, we also split the 70 experimental trials of each block into two phases of 35 trials and conducted a 2 (TIME PHASE: *first half* vs. *second half*) $\times$ 2 (PHYSICAL WEIGHT: 400 g vs. 800 g) repeated measures ANOVA of the task completion time. There was an effect of TIME PHASE, $F(1, 31) = 9.474$, $p = .004$, $\eta_p^2 = .234$, indicating that participants acted quicker in the second half ($M = 1391.51$ ms, $SD = 127.85$ ms) than in the first half ($M = 1418.33$ ms, $SD = 131.58$ ms). There

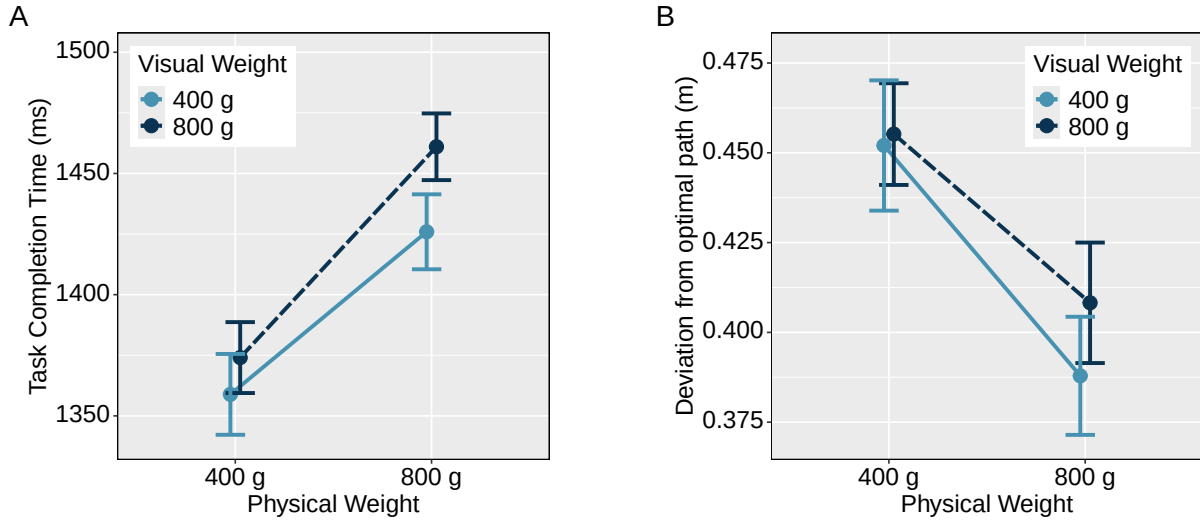


Figure 3: (A) Mean task completion times and (B) mean deviation from the optimal path. The error bars show the within-subject standard error [31].

was no interaction between PHYSICAL WEIGHT and TIME PHASE, $F(1, 31) = 1.563, p = .211, \eta_p^2 = .048$.

4.2 Precision

A 2×2 ANOVA of the deviation from the optimal path showed a main effect of PHYSICAL WEIGHT, $F(1, 31) = 7.364, p = .010, \eta_p^2 = .197$, indicating that participants performed more precise pointing movements when using the 800 g device compared to the 400 g device. We found no effect of VISUAL WEIGHT, $F(1, 31) = 0.806, p = .376, \eta_p^2 = .025$ and no interaction effect of PHYSICAL WEIGHT $\times$ VISUAL WEIGHT, $F(1, 31) = 0.314, p = .579, \eta_p^2 = .010$. The mean deviations from the optimal path are presented in Figure 3.

4.3 Questionnaires

A 2×2 ANOVA of the Borg scores revealed a main effect of PHYSICAL WEIGHT, $F(1, 31) = 42.817, p < .001, \eta_p^2 = .58$, showing that participants experienced higher exertion when using the heavier physical weight. There was no effect of VISUAL WEIGHT, $F(1, 31) = 2.491, p = .125, \eta_p^2 = .074$. and no interaction effect of PHYSICAL WEIGHT $\times$ VISUAL WEIGHT, $F(1, 31) = 0.136, p = .715, \eta_p^2 = .004$.

We totaled the questionnaire items of the Reality Judgment and Presence Questionnaire to obtain an overall realism score. There was no main effect of PHYSICAL WEIGHT, $F(1, 31) = 1.169, p = .288, \eta_p^2 = .036$, or of VISUAL WEIGHT $F(1, 31) = 0.110, p = .742, \eta_p^2 = .004$. There was also no interaction effect, $F(1, 31) = 2.437, p = .129, \eta_p^2 = .073$.

5 Discussion & Conclusion

In this study, we examined how handheld objects' physical and visually conveyed weight affect pointing movements in VR. We found, that increasing the physical weight of the pointing device in VR significantly enhanced precision while elongating the duration of the movement. Interestingly, our results also show, that

altering visual weight cues alone led the participants to adapt their movement strategies, with the visual weight of the handheld object significantly affecting movement speed. This suggests, that a virtual tool's appearance may need to be considered when designing for efficient interaction in VR, with lighter appearances potentially fostering quicker responses. Moreover, the effect of physical weight suggests that increased physical weight may enhance users' control and accuracy in goal-directed tasks and could provide a strategy for improving precision in VR. For instance, weight-adjusting controllers [19, 22, 40] could be leveraged when requiring high precision, such as when selecting closely spaced objects or interacting in cluttered environments.

Our outcome aligns with the theoretical framework for understanding speed-accuracy relations in goal-directed movements proposed by Elliott et al. [9, 10], which posits that goal-directed movements are organized to optimize speed, accuracy, and energy expenditure. According to the framework, these movements consist of an initial phase that brings the limb close to the target and a corrective phase that ensures the target is reached precisely, whereby humans prioritize energy efficiency and accuracy when the cost of the corrective phase is perceived as high. With the heavier controller, participants likely anticipated higher energy and time costs with overshooting the target and performing corrective movements. This aligns with the significantly higher rating of perceived exertion for the heavier controller. Consequently, they adopted a strategy of slower, more controlled initial movements to reduce the need for corrective movements, thereby achieving greater accuracy at the expense of task completion time.

Participants displayed slower task completion times when interacting with a visually heavier virtual object, regardless of the physical weight of the controller. Hence, from the perspective of the framework by Elliott et al. [9, 10], when objects appear visually heavier, users may subconsciously assume that they require more effort to be moved and thus anticipate higher energy consumption,

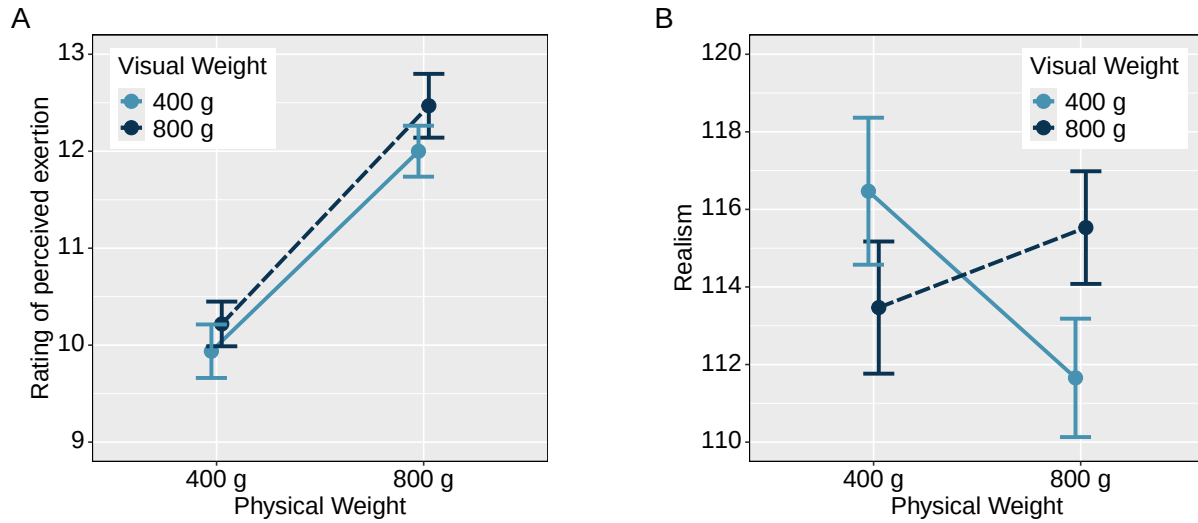


Figure 4: (A) Mean rating of perceived exertion and (B) mean realism score. The error bars show the within-subject standard error [31].

resulting in slower movement initiation and execution, despite no differences in physical weight. While the participants exhibited slower movement speeds with increasing visual weight, there was no corresponding increase in precision. This may result from the users' reliance on visual weight, which leads to planning and executing movements that are not optimized for the actual weight load. This finding might hold additional implications for the design of VR tools, as it suggests, that designing VR tools with visual representations that mirror the physical weight of the controller might optimize task execution.

Our finding aligns with research on weight illusions demonstrating that visual weight cues, such as size and material, influence the perceived object weight, even in the absence of actual weight changes [5]. Previous work demonstrated, that, in VR, these illusions can cause the heavier-looking of two equally weighted objects to be perceived as heavier [25, 36]. This aligns with our study, where the heavier appearance elicited slower responses, suggesting it was perceived as heavier. However, in the size-weight illusion, the larger of two objects is perceived as lighter, contrary to our results. The size-weight illusion procedure involves comparing the weights of novel objects with unknown weights, whereas our visual stimuli were designed to associate a particular mass (the weight of the respective physical pointing device) with them. As visual identification of a familiar object alone can trigger appropriate motor planning [13], this can lead to differences in force output. Additionally, size variations of the stimuli in our experiment also changed their (expected) moment of inertia. Due to these discrepancies regarding effect and procedure, we do not assume that the traditional size-weight illusion was the underlying phenomenon. Instead, we interpret the slower movements with the heavier visual appearance as a strategy of relying more dominantly on the mental link between the device's appearance and physical weight, in line with the Bayesian decision theory in sensorimotor control [24]. This theory posits that with increasing uncertainty in the sensory input,

the central nervous system increases the weighting of the prior knowledge when planning actions. Thus, in the case of our study, the central nervous system may have increased the weighting of the stable prior information about the heavy device's weight and decreased the weighting of the conflicting sensory weight cues.

Although physical and visual weight were effective in modulating task performance, they did not affect the realism scores. As related work has demonstrated the value of congruence of physical and visual weight for realism [19, 22, 40], the absence of an effect in the present study could arise from the rather abstract design of the devices lacking intuitive realism. Furthermore, we cannot rule out that realism was indeed affected and that our experiments' data did not provide sufficient power to detect such effect (see Figure 4 B). Future studies could consider testing more realistic representations of weight, to explore how to optimize task performance through weight manipulation without negative realism.

The study employed a priming task to establish consistent initial expectations and sensorimotor knowledge of the objects' weights. While this step was critical for ensuring a controlled evaluation of physical and visual weight effects, it remains unclear whether the effect of visual weight observed in this study would persist when participants rely on less recent weight associations. Furthermore, the priming task might have added fatigue, potentially influencing participants' performance during the main task. However, our analysis found no interaction between physical weight and time phase regarding task completion times, suggesting fatigue over time did not play a role in shaping task performance. To validate the application of our findings in VR scenarios that do not involve immediate prior sensorimotor experience, future research should explore whether the effects are robust in the absence of a priming task.

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